

Exam 1

ECE 2201: Circuit Analysis I

Fall 2024

October 5, 2024

1. This exam is closed book, closed notes and a crib sheet has been provided to you with the exam. Print your name below.
2. Show all work on these pages. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution which is not given in a reasonable order will lose credit.
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 90 minutes to work on this exam.

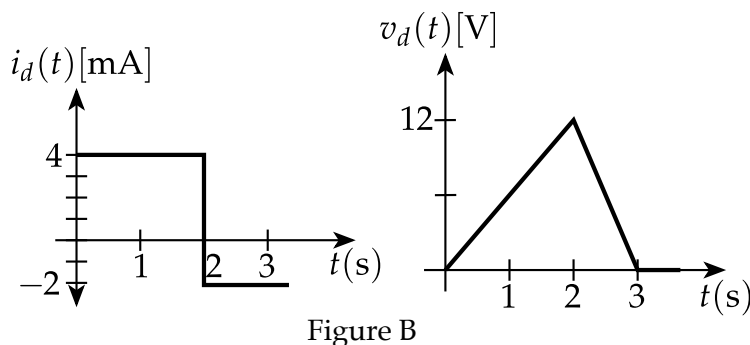
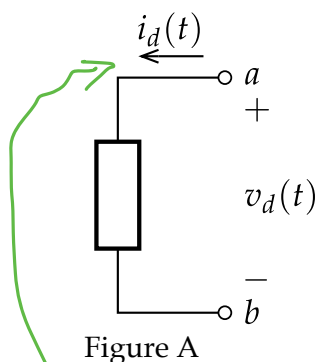
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Problem 1

A Device shown in Figure A has a current, $i_d(t)$, and a voltage, $v_d(t)$, as shown in Figure B.

- Write, in equation form, the power delivered by the device for $0 \leq t \leq 3[s]$.
- Calculate the energy delivered by the device for time $1[s] \leq t \leq 3[s]$.



$$i_d(t) = \begin{cases} 4[\text{mA}] & 0 \leq t \leq 2[\text{s}] \\ -2[\text{mA}] & 2[\text{s}] < t \leq 3[\text{s}] \end{cases} \quad v_d(t) = \begin{cases} 6[\frac{\text{V}}{\text{s}}]t & 0 \leq t \leq 2[\text{s}] \\ -12[\frac{\text{V}}{\text{s}}]t + 36[\text{V}] & 2[\text{s}] < t \leq 3[\text{s}] \end{cases}$$

(a) $p = -i_d(t) \cdot v_d(t) = \begin{cases} -24[\frac{\text{mW}}{\text{s}}] \cdot t & 0 \leq t \leq 2[\text{s}] \\ -24[\frac{\text{mW}}{\text{s}}]t + 72[\text{mW}] & 2[\text{s}] < t \leq 3[\text{s}] \end{cases}$

↑
b/c current flows into device - power delivered is negative

(b) $\frac{dw}{dt} = p \Rightarrow dw = p dt \Rightarrow w = \int p dt$

$$w = \int_{1[\text{s}]}^{2[\text{s}]} -24[\frac{\text{mW}}{\text{s}}] \cdot t dt + \int_{2[\text{s}]}^{3[\text{s}]} (-24[\frac{\text{mW}}{\text{s}}]t + 72[\text{mW}]) dt$$

$$= -12[\frac{\text{mW}}{\text{s}}] \cdot t^2 \Big|_{1[\text{s}]}^{2[\text{s}]} - 12[\frac{\text{mW}}{\text{s}}] t^2 \Big|_{2[\text{s}]}^{3[\text{s}]} + 72[\text{mW}] \cdot t \Big|_{2[\text{s}]}^{3[\text{s}]}$$

note:
[mW·s] = [mJ]

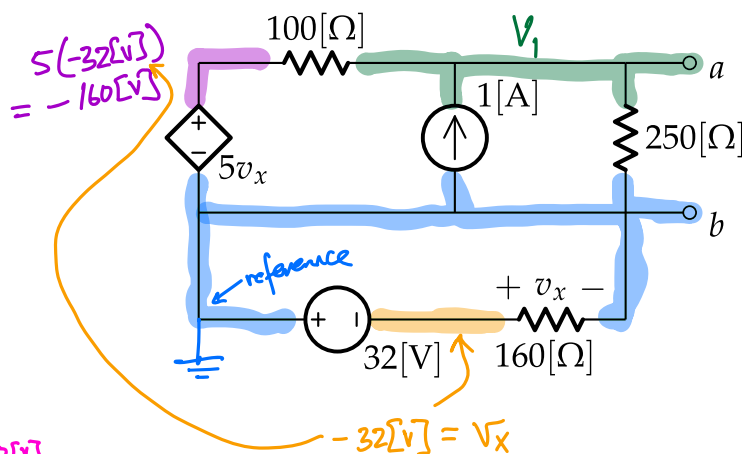
$$= -12 \cdot 4 [\text{mJ}] + 12 \cdot 1 [\text{mJ}] - 12 \cdot 9 [\text{mJ}] + 12 \cdot 4 [\text{mJ}] + 72 \cdot 3 [\text{mJ}] - 72 \cdot 2 [\text{mJ}]$$

$$= -24 [\text{mJ}]$$

Problem 2

Refer to the circuit shown below.

- Find the Thevenin equivalent circuit with respect to nodes a and b. Include clear labels and units in your intermediate work. Draw your final Thevenin equivalent circuit in the box below with clear labels and units.
- Find the Norton equivalent circuit with respect to nodes a and b and draw your final solution in the box below with clear labels and units.



$$1[A] = \frac{V_1 - (-160[V])}{100[\Omega]} + \frac{V_1}{250[\Omega]}$$

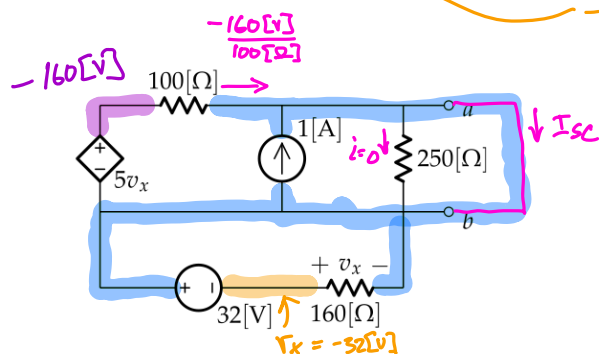
mult. by $500[\Omega]$

$$500[V] = 5V_1 + 800[V] + 2V_1$$

$$7V_1 = -300[V]$$

$$V_1 = -\frac{300}{7}[V] = V_{oc}$$

$$= -42.9[V]$$

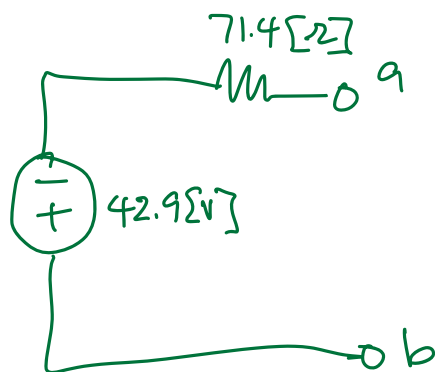


$$I_{sc} = \frac{-160[V]}{100[\Omega]} + 1[A]$$

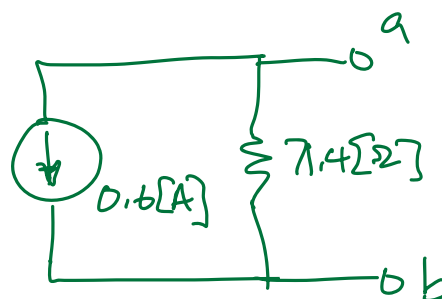
$$= -0.6[A]$$

$$\therefore R_{TH} = R_N = \frac{V_{oc}}{I_{sc}} = \frac{-\frac{300}{7}[V]}{-0.6[A]} = 71.4[\Omega]$$

Thevenin Circuit

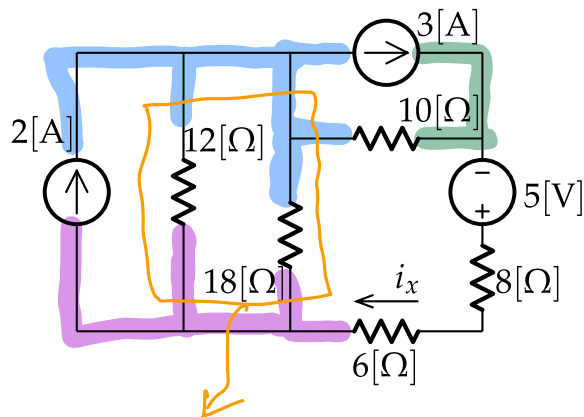


Norton Circuit



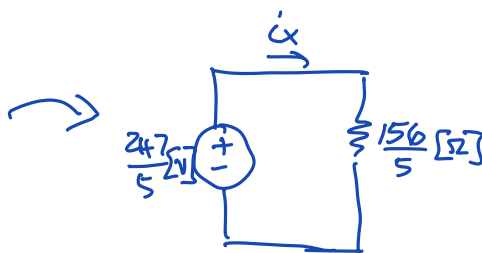
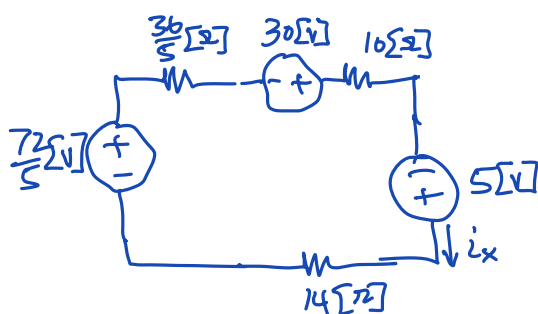
Problem 3

Use source transformations and series/parallel combinations to simplify the circuit and find the current i_x .



$$12[\Omega] \parallel 18[\Omega]$$

$$= \frac{12 \cdot 18}{12 + 18} [\Omega] = \frac{12 \cdot 18}{30} = \frac{36}{5} [\Omega] = 7.2 [\Omega]$$



$$\left(\frac{36}{5} + \frac{50}{5} + \frac{70}{5} \right) [\Omega]$$

$$= \frac{156}{5} [\Omega]$$

$$= 31.2 [\Omega]$$

$$\left(\frac{72}{5} + \frac{150}{5} + \frac{25}{5} \right) [V]$$

$$= \frac{247}{5} [V] = 49.4 [V]$$

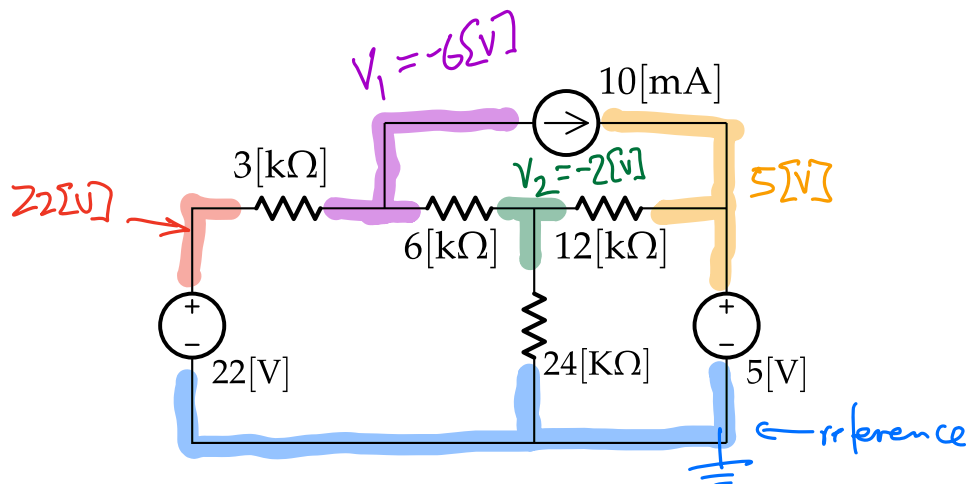
$$i_x = \frac{\frac{247}{5} [V]}{\frac{156}{5} [\Omega]} = \frac{247}{156} [A] = 1.58 [A]$$

$$i_x = \underline{1.58 [A]}$$

Problem 4

Using nodal analysis (i.e., node voltage method),

- Label two nodes needed to solve this problem as V_1 and V_2 and write the two node voltage equations – be sure to simplify the expressions. Include your answers in the box below.
- Solve for V_1 and V_2 . Include your answers in the box below.
- Determine the power p supplied by the current source. Include your answer in the box below.



Eq. 1	$3V_1 - V_2 = 16 \text{ [V]}$
Eq. 2	$-4V_1 + 7V_2 = 10 \text{ [V]}$
V_1	-6 [V]
V_2	-2 [V]
p	110 [mW]

$$\frac{V_1 - 22 \text{ [V]}}{3 \text{ [k}\Omega]} + \frac{V_1 - V_2}{6 \text{ [k}\Omega]} + 10 \text{ [mA]} = 0$$

mult. by $6 \text{ [k}\Omega]$

$$2V_1 - 44 \text{ [V]} + V_1 - V_2 + 60 \text{ [V]} = 0$$

$$3V_1 - V_2 = -16 \text{ [V]}$$

$$V_2 = 3V_1 + 16 \text{ [V]}$$

$$\frac{V_2 - V_1}{6 \text{ [k}\Omega]} + \frac{V_2}{24 \text{ [k}\Omega]} + \frac{V_2 - 5 \text{ [V]}}{12 \text{ [k}\Omega]} = 0$$

mult. by $24 \text{ [k}\Omega]$

$$4V_2 - 4V_1 + V_2 + 2V_2 - 10 \text{ [V]} = 0$$

$$-4V_1 + 7V_2 = 10 \text{ [V]}$$

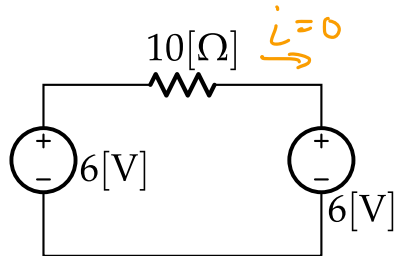
$$\begin{aligned} -4V_1 + 7(3V_1 + 16[V]) &= 10[V] \\ 17V_1 &= 102[V] \end{aligned} \quad \boxed{V_1 = -6[V]}$$

$$\begin{aligned} V_2 &= 3(-6[V]) + 16[V] \\ \boxed{V_2 &= -2[V]} \end{aligned}$$

$$\begin{aligned} P &= i \cdot V = 10[{\text{mA}}](5[V] - V_1) \\ &= 10[{\text{mA}}](5[V] - (-6[V])) \\ &= 10[{\text{mA}}](11[V]) = 110[{\text{mW}}] \end{aligned}$$

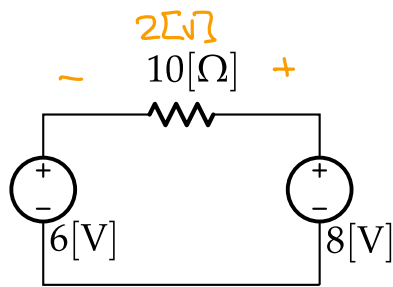
Problem 5

For each of the following circuits, find the power p delivered to the $10[\Omega]$ resistor.



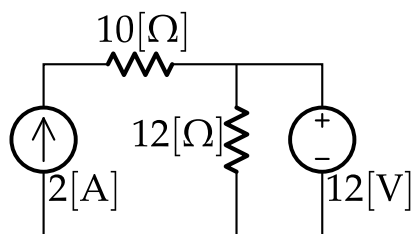
$$p = i^2 R = 0$$

$$p = \underline{0}$$



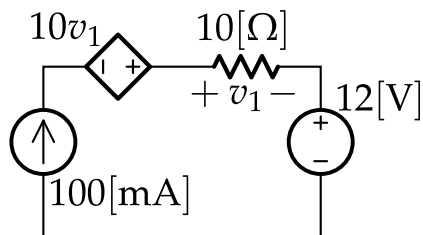
$$p = \frac{v^2}{R} = \frac{(2[V])^2}{10[\Omega]}$$

$$p = \underline{400[mW]}$$



$$p = i^2 R = (2[A])^2 \cdot 10[\Omega]$$

$$p = \underline{40[W]}$$



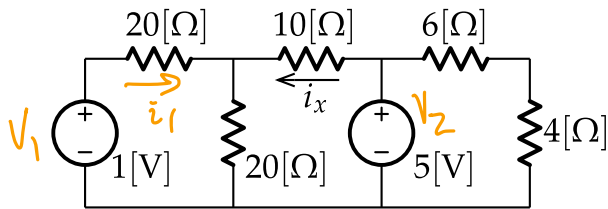
$$p = i^2 R = (100[mA])^2 \cdot 10[\Omega]$$

$$p = \underline{100[mW]}$$

$$= (0.1[A])^2 \cdot 10[\Omega]$$

$$0.1[W]$$

Use superposition to determine the current flowing through the $10[\Omega]$ resistor, i_x , and then determine the power p delivered to the resistor.



$$i_x = \underline{0.225 [A]}$$

$$p = \underline{506 [\mu W]}$$

when $V_2 = 0$

$$i_x = \frac{5[V]}{10[\Omega] + 10[\Omega]} = 0.25 [A]$$

$$(0.225 [A])^2 \cdot 10[\Omega] = 0.506 [W]$$

when $V_1 = 0$

$$i_1 = \frac{1[V]}{20[\Omega] + 20[\Omega] \parallel 10[\Omega]}$$

$$= 0.0375 [A]$$

$$i_x = - \frac{i_1 (10[\Omega] \parallel 20[\Omega])}{10[\Omega]}$$

$$= -0.025 [A]$$

